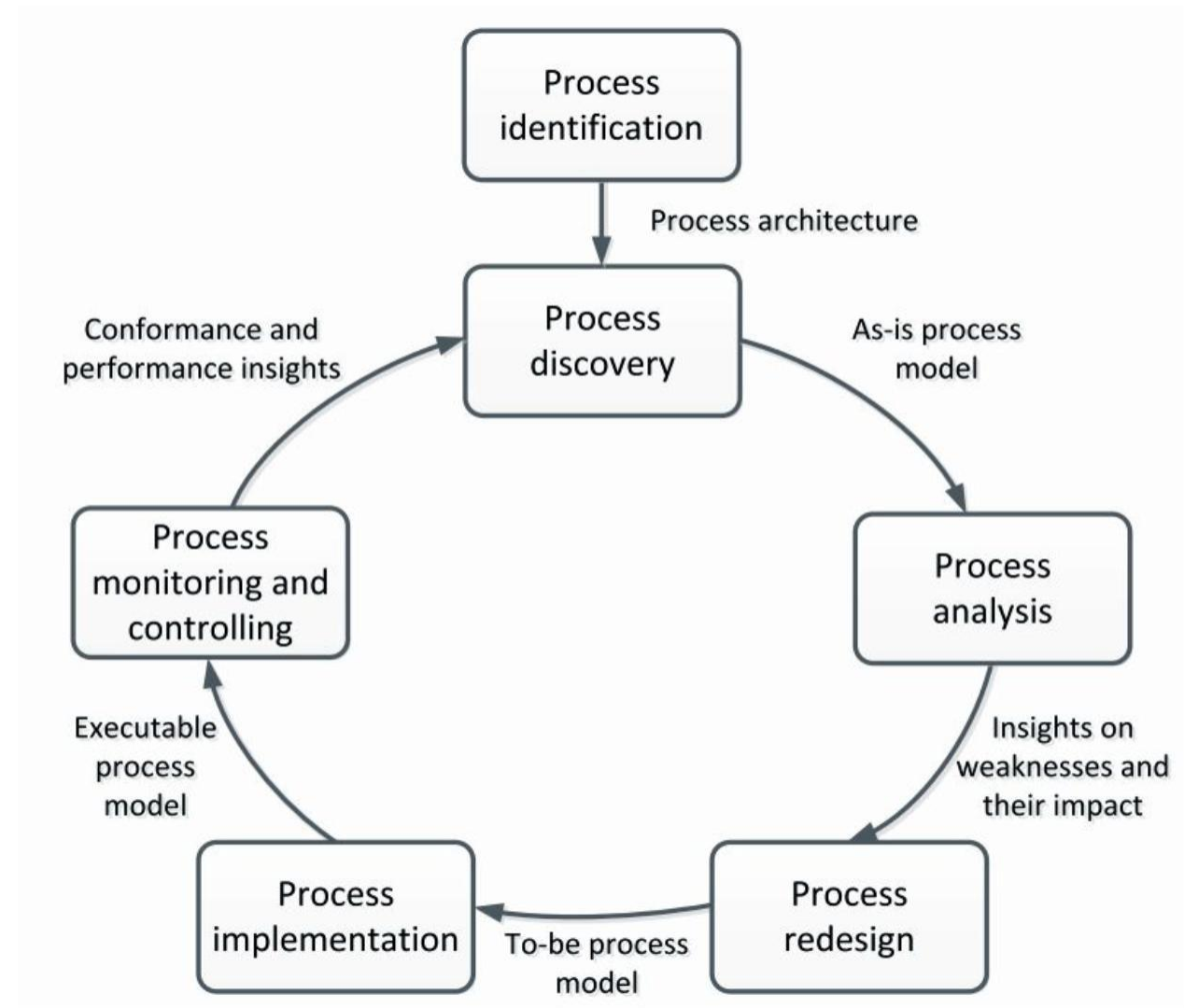


BUSINESS PROCESS MANAGEMENT

IS 2206

Lecture 03

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OUTLINE

1. Phases of Process Identification
2. Designing of Process Architecture
3. Relationships in Process Architecture

TOPIC 01

Phases of Process Identification



1.1 What is Process Identification

- A collection of activities aimed at systematically **defining** a company's set of business processes and establishing explicit criteria for **prioritizing** them is known as process identification.
- A **process architecture**, which reflects the business processes and their interrelationships, is the result of process identification.
- A process architecture is a framework for prioritizing and specifying the scope of process modeling and redesign projects.

1.2 Focusing on Key Processes

- *What processes are executed in the organization? and Which ones should the organization focus on? Does the process meet its goals? What aspects of the process need improvement? What changes would improve those aspects of the process?*
- Some processes need to receive priority because they are of strategic importance to an organization's survival.
- Other processes might show striking problems, which should be resolved for the sake of all involved stakeholders.

1.3 Different Types of Processes

Management processes involve monitoring primary and secondary processes to ensure that the organization is meeting overall financial and operational goals.

**Management
Processes**

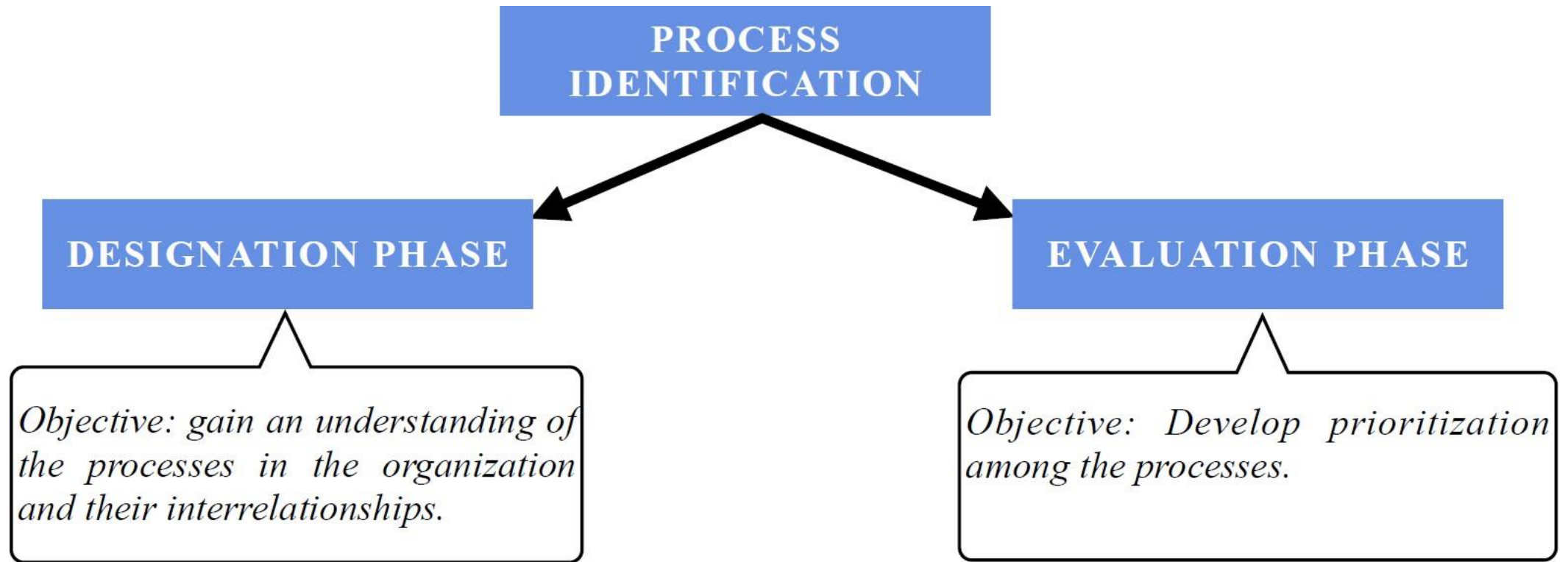
Core business processes directly deliver values to the organization and its customers.

**Core
Processes**

Support business processes provide value to the internal customers and do not generate income in themselves.

**Support
Processes**

1.4 Key Phases of Process Identification



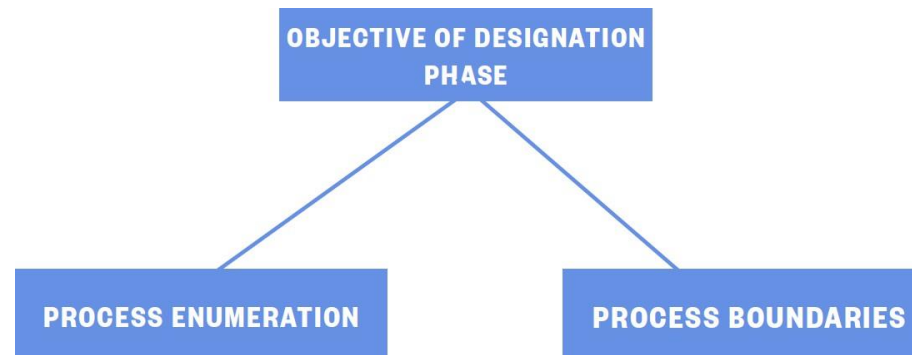
1.4.1 Designation Phase

Process Enumeration

- The typical number of processes are unclear.
- Represent a Trade-off between:
 - ✓ (Ensure process scope is) **manageable**
 - ✓ (Process scope determines the) **potential impact** on when developing a BPM solution.

Process Boundaries

- Processes are interdependent
- If the process is large, it cannot be easily managed, in terms of scope , hierarchical chain and speed of action.
- Processes change over time
 - ✓ Identification should be exploratory and iterative



1.4.1 Designation Phase (Continued)

*Davenport has suggested that it may be useful to identify both **broad and narrow processes** to balance the advantages and disadvantages of a large process scope.*

Broad Processes

- Processes that the organization feels it's important to completely overhaul the existing operations at some point.

Narrow Processes

- The processes that are not targeted for major overhauls.
- These processes need to be actively monitored.
- Subject to continuous fine-tuning and updating.

1.4.2 Evaluation Phase

Objective: Develop prioritization among the processes to facilitate process management activities, such as modelling, redesign, automation, monitoring etc.

Process Evaluation Criteria

- **Importance:** Assess the **strategic relevance** of each process. The goal is to find out which process has the greatest impact on the company's strategic goals.
- **Dysfunction:** Render the high-level judgment of the “**health**” of each of the processes. Determine which processes are in the deepest trouble.
- **Feasibility:** Each process should be determined how susceptible they are to process management initiatives, incidental or on a continuous basis.

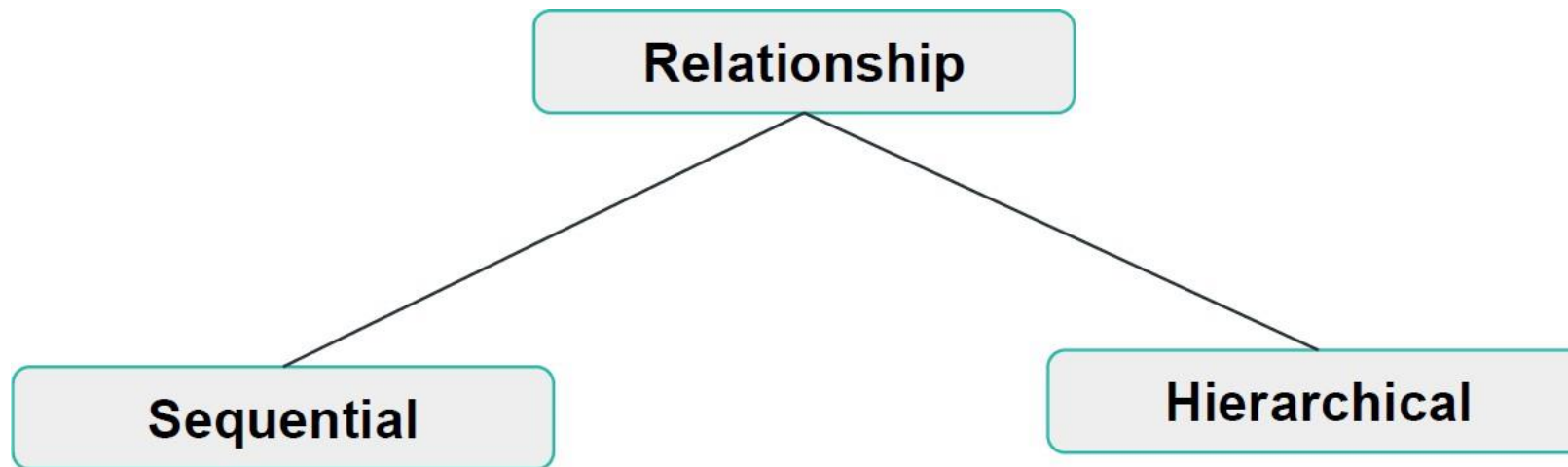
TOPIC 02

Designing of Process Architecture



2.1 Designing a Process Architecture

- Process architecture is a **conceptual model** that shows the processes of a company and makes their relationships explicit.
- Process Architecture is very useful to gain a common understanding of the business processes in the organization.



2.1.1 Sequential Relationship

- One process provides an output that the other process takes it as an input.
- Normally we can consider the consumer-producer relationship in two different perspectives.
- One is the narrow interpretation. Another one is the broad interpretation.

Quote to Order Process → The Order (*Output*)

The Order (*Input*) → Order to Cash Process

2.1.2 Hierarchical Relationship

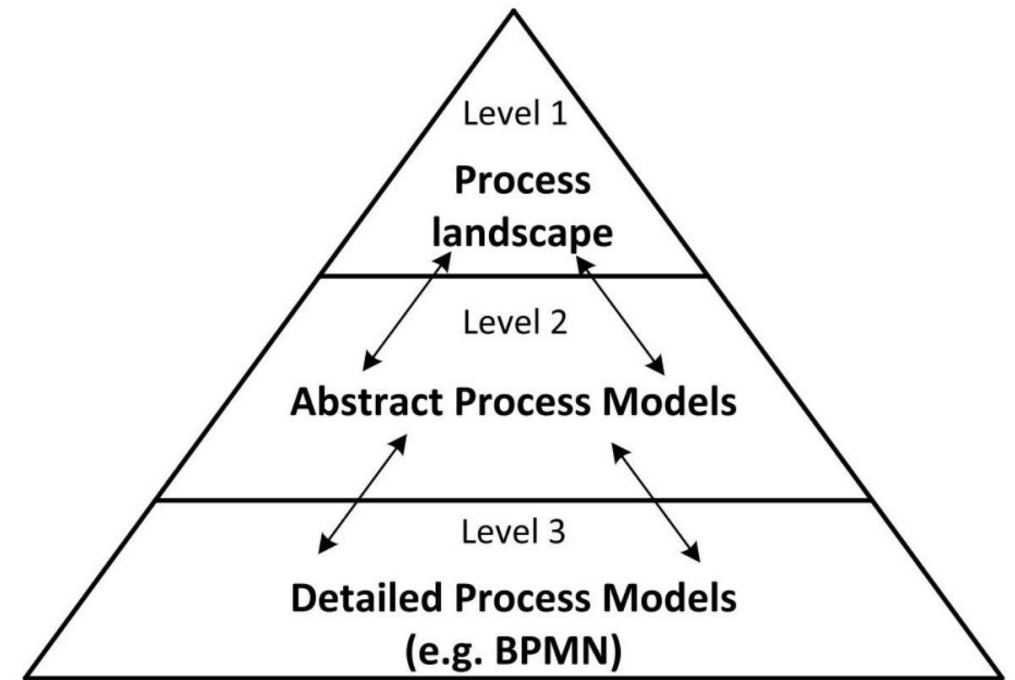
- A hierarchical decomposition from core business processes to the task level.

Example

- A broad process like order management can be related to the more narrowly defined processes of order booking, billing, shipment and delivery. All these can be considered as sub processes of order management.

2.2 Process Architecture

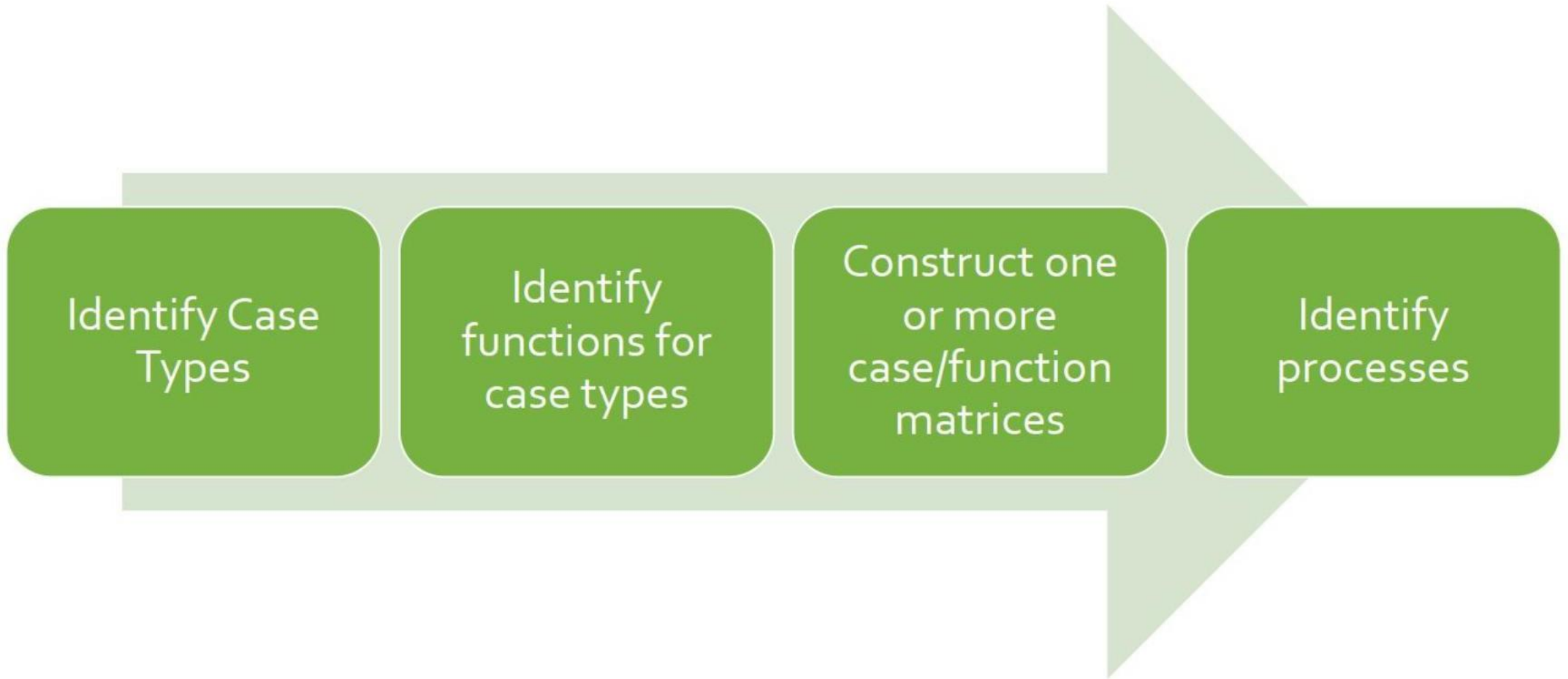
- **Level 1:** It shows the main processes on a very abstract level. Each of the elements of the process landscape model points to a more concrete business processes on level two.
- **Level 2:** This level two shows the processes at a finer degree of granularity, but still in a quite abstract way. Each element on level two points further to a process model on level three.
- **Level 3:** The process models on this third level show the detail of the processes including control flow, data inputs and outputs, and assignment of participants



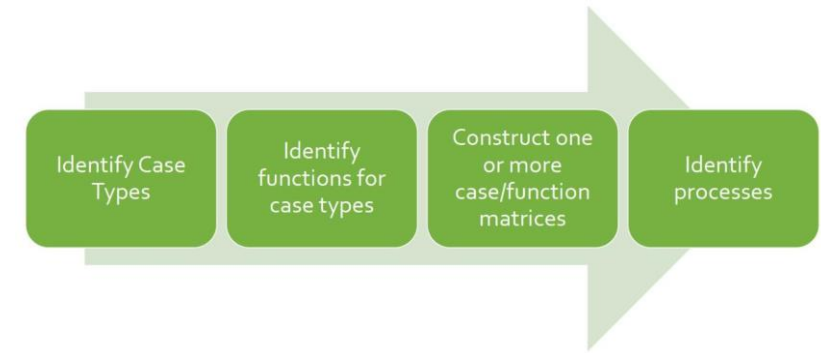
2.3 Benefits of Process Architecture

- A full overview of all activities and how they relate to one another.
- Aligning regular activities with overall business strategies and goals.
- Identifying opportunities for automation.
- Predicting the impact of systems changes.
- When analyzing the process architecture, bottlenecks and resulting delays become apparent

2.4 Dijkman Approach



2.4.1 Identify Case Types



- A case is something that an organization (or a part of it) handles.
- Typically, it is a **product or a service** that is delivered by the organization to its customers.
 - Ex: Insurance (service), Toy (product)
- Cases can also refer to products or services delivered by one department to another in the same organization.
- The cases can be classified using any number of properties.
 - Ex: In an insurance company, according to the
 - service type - home insurance, car insurance. etc.
 - channel type - online, telephone, etc.
 - both type - home insurance via telephone

2.4.1 Identify Case Types (Continued..)

- Commonly used Types of Cases

- ❖ **Product type**

- Types of products handled by the organization. They are usually hierarchically decomposed.

- ❖ **Service type**

- If an organization/part of an organization handles services this property identifies the services handled.

- ❖ **Channel type**

- Channels through which the organization contacts its customers Ex: face-to-face, Online

- ❖ **Customer type**

- Types of customers the organization is dealing with.

- Ex: Airline company – frequent flyers and regular customers.

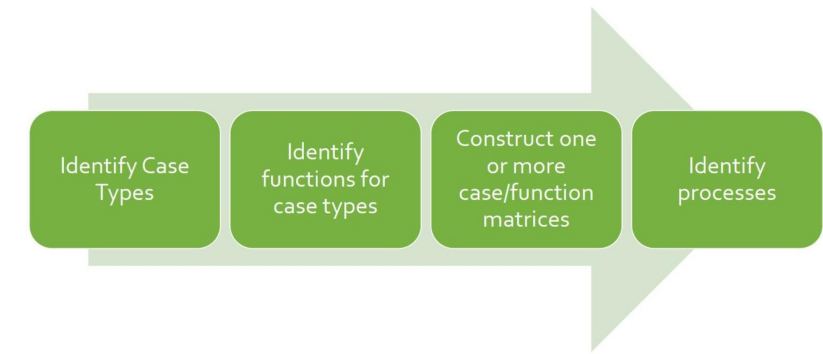
2.4.1 Identify Case Types (Continued..)

- Case Properties
 - ❖ Any property that has the same characteristics but handled differently can be identified as separate property.
 - ❖ When identifying case types properties that distinguish cases should be selected.
 - ❖ However, the properties that distinguish cases but do not lead to different behaviors should not be included.

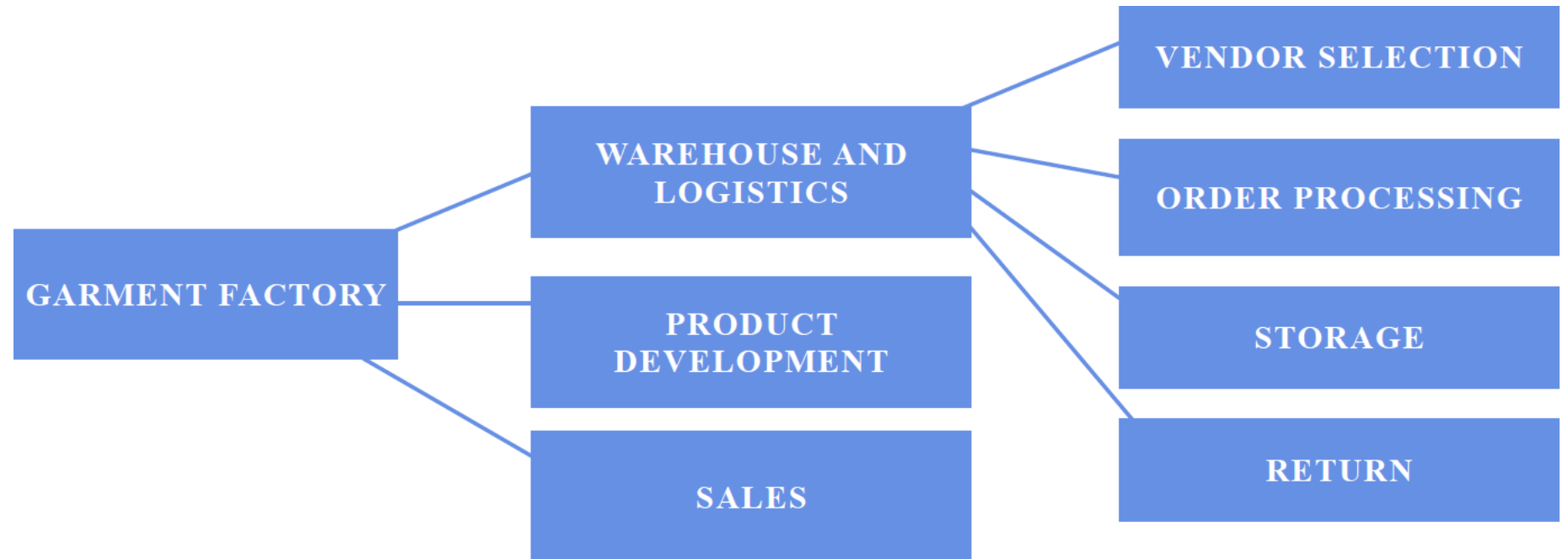
Ex:

- ✓ Retail shop – same process for all products sold, so product type is not important.
- ✓ Hotels– different products (rooms, foods, wedding halls sell in different ways) so product type is important.

2.4.2 Identify Functions for Case Types



- The function dimension classifies the functions of the organization.
- A function is something the organization does.



2.4.2 Identify Functions for Case Types (Continued..)

- To identify functions in an organization it requires interviews with different people in the organization.
- *Interviews must be held with different process participants in the organization.*
- These interviews could either directly identify the functions or to check which extent the functions from a reference model apply to the organization.
- Functions that are performed in an organization can be related to existing classifications that are proposed by reference models.
- Such reference models can serve as a starting point to develop a classification of business functions. Such as ITIL, PCF, APQC.

2.4.2 Identify Functions for Case Types (Continued..)

- **Functional Decomposition**

Hierarchical decomposition of functions can be made.

business function	Europe	
	purchasing	sourcing
		order-to-pay
	sales	marketing
		operational sales

business function	North America	
	sourcing	
	marketing	
	order handling	

2.4.3 Identify Case/Function Matrices

Identify Case
Types

Identify
functions for
case types

Construct one
or more
case/function
matrices

Identify
processes

- Identification of cases and functions lead to a matrix that has different case types as columns and different functions as rows.
- If the corresponding function can be performed by the corresponding case the cell matrix contains a 'X'.
- Example 1:

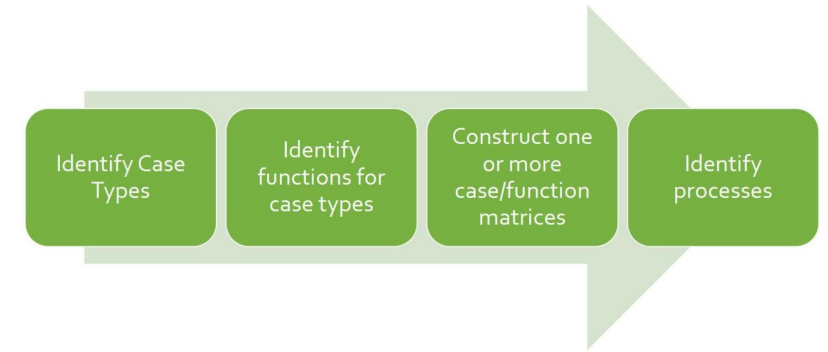
		Private Customers	Corporate Customers	Internal Customers
Management	Process			X
	Line			X
	Project			X
Operations	Savings	X	X	
	Loans	X	X	
	Checking	X	X	
Support	HRM			X
	ICT			X
	Finance			X
	Marketing			X

2.4.3 Identify Case/Function Matrices (Continued..)

- Example 2:
A process architecture for the harbor authority

			case type			
			Sea	Road	Rail	Inland
business function	pre-arrival	notify ETA	Inbound Planning			
		notify authorities				
		reserve tow-boat				
	arrival		Inbound Handling			
	trans-shipment	stacking/handling		Outbound Handling		
		payment				
	departure	infrastructure info				
		notify ETD				

2.4.4 Identify Processes



- Determine which combinations of business functions and case types form into a business process.
- *Entire matrix forms into one big process. Or a single cross line in the matrix forms a process.*
- We establish this trade-off by the use of eight guidelines.
- When a guideline applies, this may lead to a separation of processes between rows (a vertical split) or to a separation of processes between columns (a horizontal split).
Ex: Guidelines 1 to 4 lead to a vertical split. Guidelines 5 to 8 lead to a horizontal split.
- Note that the guidelines are not absolute: they may or may not apply to a particular organization and they are not the only rules that should be considered in specific cases.

2.4.4 Identify Processes (Continued..)

- Example: A case/function matrix for a mortgage broker

		Case type			
		Netherlands		Belgium	
		Composite	Simplex	Composite	Simplex
Risk Management	Product Risk Assessment	X	X	X	
	Client risk Assessment	X	X	X	
Mortgage Brokering	Selecting	X		X	
	Offering	X		X	
	Contracting	X	X	X	
Finance	Payment	X	X	X	
	Collection	X	X	X	
Product Development		X		X	

2.4.4 Identify Processes – Guideline 1

If a process has different flow objects, it can be split up vertically.

Flow Objects

- A flow object is an object in the organization that flows through a business process
- Typically, each business process has a single flow object, such that flow objects can be used to identify business processes.
- Consequently, if multiple flow objects can be identified in a business process, this is a strong indication that the process should be split up.
- **Ex: Receiving the Mortgage application (Flow Object) is the starting point to initiate Mortgage Brokering Process.**

		case type			
		Netherlands		Belgium	
		Composite	Simplex	Composite	Simplex
risk management	product risk assessment	Product Development and Assessment			
	client risk assessment	X	X	X	
mortgage brokering	selecting	X		X	
	offering	X		X	
	contracting	X	X	X	
finance	payment	X	X	X	
	collection	X	X	X	
product development		Product Development and Assessment			

2.4.4 Identify Processes – Guideline 2

If the flow object of a process changes multiplicity, the process can be split up vertically.

Multiplicity Flow Objects

- In a business process a single flow object is sometimes used, while at other times multiple flow objects of the same type are used.
- This is typical for batch processing, in which certain activities are performed for multiple customer cases in batch at the same time.
- In the same process, the number of flow objects that are processed per activity differs this may be a reason for splitting up the process.
- **Ex: Mortgage application process has single flow object (mortgage application) and payment collection happen on all mortgages. Collection of payments for all mortgages happens as a batch at the end of each month.**

2.4.4 Identify Processes – Guideline 3

If a process changes transactional state, it can be split up vertically.

Transaction State

- A business process goes through a number of transactional states.
- Ex: the initiation, the negotiation, the execution and the acceptance state.
- A transition in a process from one state to another state is an indication that the process can be split up.
- Ex: **Mortgage Selection process, Contract Signing Process and Mortgage Payment process** happen for different transactional states.

		Case type			
		Netherlands		Belgium	
		Composite	Simplex	Composite	Simplex
Risk Management	Product Risk Assessment	Product Development			
	Client risk Assessment	X	X	X	
Mortgage Brokering	Selecting	X		X	
	Offering	X		X	
	Contracting	X	X	X	
Finance	Payment	X	X	X	
	Collection	X	X	X	
Product Development		Product Development			

2.4.4 Identify Processes – Guideline 4

If a process contains a logical separation in time, it can be split up vertically.

Logical Separation of Time

- A process contains a logical separation in time, if its parts are performed at different time intervals.
- Intervals that can typically be distinguished include: once per customer request, once per day, once per month and once per year.
- **Ex: Payment collection happens once per month.**

		Case type			
		Netherlands		Belgium	
		Composite	Simplex	Composite	Simplex
Risk Management	Product Risk Assessment	Product Development			
	Client risk Assessment	X	X	X	
Mortgage Brokering	Selecting	X		X	
	Offering	X		X	
	Contracting	X	X	X	
Finance	Payment	Mortgage Payment			
	Collection	Mortgage Collection			
Product Development		Product Development			

2.4.4 Identify Processes – Guideline 5

If a process contains a logical separation in space, it can be split up horizontally.

Logical Separation is Space

- A process contains a logical separation in space, if it is performed at multiple locations and is performed differently at those locations.
- **Ex: Risk Assessment, Mortgage Brokering and Product Development may differ between the Netherlands and Belgium, due to country-specific rules and regulations.**

(diagram change to split Risk Assessment and Product Development)

Here it has considered application processing done in two locations.

		Case type			
		Netherlands		Belgium	
		Composite	Simplex	Composite	Simplex
Risk Management	Product Risk Assessment	Product Development			
	Client risk Assessment	X	X	X	
Mortgage Brokering	Selecting	Mortgage Application		Mortgage Application	
	Offering	X NL		X BE	
	Contracting	X	X	X	
Finance	Payment	Mortgage Payment			
	Collection	Mortgage Collection			
Product Development		Product Development			

2.4.4 Identify Processes – Guideline 6

If a process contains a logical separation in another relevant dimension, it can be split up horizontally.

Logical Separation in other Dimensions

- Like with the separation in space, it is not sufficient for processes to just be separated.
- The separation must be such that there is no choice but to perform the processes differently for the different logical units.
- Ex: Mortgage applications for simplex products in the Netherlands will be separated by another logical unit.

2.4.4 Identify Processes – Guideline 7

If a process is split up in a reference model, it can be split up.

Reference Models

- A reference process architecture is an existing process architecture that is pre-defined as a best-practice solution.
- **Ex: if a reference financial services process architecture exists, its structure can be used as an example or starting point to structure your own process architecture.**

		case type			
		Netherlands		Belgium	
		Composite	Simplex	Composite	Simplex
risk management	product risk assessment	X PD NL X		PD BE	
	client risk assessment	X	X	X	
mortgage brokering	selecting	Mortgage Application NL		Mortgage Application BE	
	offering	X		BE	
	contracting	X	X	X	
finance	payment	X Mortgage payment X		X	
	collection	X Mortgage Collection X		X	
product development		PD NL		PD BE	

2.4.4 Identify Processes – Guideline 8

If a process covers (many) more functions in one case type than in another, it can be split up horizontally.

Function Case

- The application of this last rule depends upon the current decomposition of processes.
- **Ex: whether a process has many more crosses in one column than in another. If so, this is a strong indication that the process should be split up for these two case types.**

		case type			
		Netherlands		Belgium	
		Composite	Simplex	Composite	Simplex
risk management	product risk assessment	X PD NL X		PD X BE	
	client risk assessment	X Composite	X Simplex	X Mortgage Application	
mortgage brokering	selecting	X Mortgage Application NL	X Mortgage Application NL	X Mortgage Application BE	
	offering				
	contracting	X	X	X	
finance	payment	X Mortgage Payment	X Mortgage Payment	X	
	collection	X Mortgage Collection	X Mortgage Collection	X	
product development		PD X NL		PD X BE	

TOPIC 03

Relationships in Process Architecture



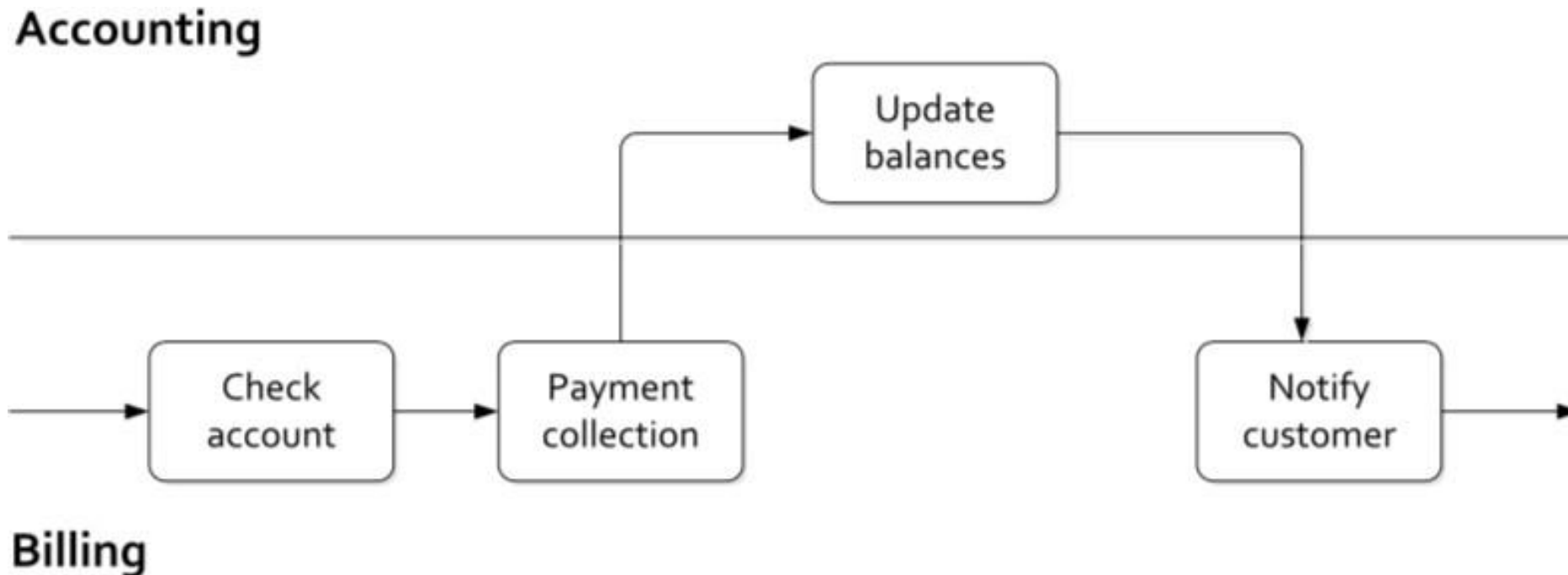
Complete the Process Architecture

- Matrix only depicts the level 1 – abstract process
- Differs in terms of case and functions they cover
- The consumer-producer relationships between the processes and the level 2,3 of the pyramid are missing
- To extend it to level 2 (process map), we should consider the various **steps that are taken within each process** and the **organizational units** that are involved in carrying out those steps.

Consumer	Producer
Mortgage Payment	Composite Mortgage Application NL
Mortgage Payment	Simplex Mortgage Application NL
Mortgage Payment	Mortgage Application BE

Example

- It is common to refer to a model on the second level of the 'pyramid' as a process map.
- The process map for the mortgage payment process



Complete the Process Architecture

- Even a process map (we are not discussing it in details) can still be said to provide an abstract view on a process.
- It commonly shows linear progress along with the various steps in a process map.
- But alternative paths, potential exceptions, iterations, etc. are all left out.
- In level 3, we have to add these deep level information to the process map.
- Process maps turn into providing a deeper level of insight into the business process by adding alternative paths, control flows, exceptions etc.

Thank You!